FairForge-2 tables: current draft

CES-FL project, updated 8 Nov 2026 (L_A, L_H-PS, RF-S, RF-SJg added) from verified run aggregates

Status of the numbers. All results are on FairForge-2 **SELECT (validation), not TEST**: seed 17, one run per method, threshold fixed on SELECT reals for 10% FPR (achieved 9.97%). TEST stays sealed until the final model, checkpoint and threshold are frozen; the paper version will report TEST (87,564 images, plus the 16,953-image unseen-generator subset) as mean ± std over 3 seeds. Grey rows are not yet available. Bold = best among the filled rows. Age and skin bins are machine estimates (InsightFace age, ITA skin tone), not human labels. CES-FL P/A/B are controls of the CES-FL recipe; L_A (6 Nov) is the matched BCE control for L_H-PS (plain linear head, encoder from FF2 BCE50, 10 epochs, selected epoch 2); L_H-PS (7 Nov) is our fairness method on the same recipe (selected epoch 10). Pre-declared rule: L_H-PS is fairer than L_A only if gender×race F_{FPR} drops by ≥2 points with AUC loss ≤0.5 and FNR ≤2×: not met (37.37→36.23); RF-S (8 Nov stretch experiment, selected by min F_{FPR} subject to AUC/FNR limits) is the only feasible run of that experiment; its JPEG-augmented variants reached lower F_{FPR} (21–24) only with AUC 99.1–99.3 and FNR 1.1–1.5%, so they have no feasible checkpoint; the full counterfactual CES-FL objective has not run on FF2.

Table 1: FairForge-2 public split: real / fake images with valid group labels, per gender×race group and split, and real images per source dataset. "Others" (Indian, Latino/Hispanic, Middle Eastern) exists only in FairFace.

Group	TRAIN	SELECT	CAL	TEST	FFHQ reals	FairFace reals
Female-Asian	1,333 / 17,079	202 / 2,493	194 / 2,479	449 / 8,711	526	1,652
Female-Black	1,524 / 18,463	194 / 2,272	205 / 2,469	475 / 8,392	793	1,605
Female-White	1,774 / 21,658	257 / 2,966	232 / 2,817	567 / 9,348	824	2,006
Female-Others	2,152 / 24,286	348 / 3,882	358 / 4,033	691 / 7,961	0	3,549
Male-Asian	2,219 / 26,500	322 / 3,833	324 / 3,950	722 / 11,573	726	2,861
Male-Black	1,795 / 23,407	238 / 3,072	248 / 3,148	585 / 10,261	886	1,980
Male-White	1,809 / 23,236	256 / 3,178	243 / 2,916	612 / 10,654	886	2,034
Male-Others	4,171 / 46,860	621 / 6,973	705 / 7,934	1,340 / 15,223	0	6,837
All	16,777 / 201,489	2,438 / 28,669	2,509 / 29,746	5,441 / 82,123	4,641	22,524

Reading the tables. Ding-style GDA has the lowest F_{FPR} on gender, race, gender×race and age, but the lowest AUC and a 3–5× higher miss rate. No CES-FL control is best on any F_{FPR} column. Most of the pooled race and age disparity comes from the source dataset: FFHQ reals are flagged as fake far more often than FairFace reals (Table 3, within-source columns; README T9 in the results branch).

Table 2: Main table in the format of Ding et al. (values ×100). g = gender (2 groups), r = race (4), i = gender×race (8). $F_{\text{FPR}} = \sum_g |\text{FPR}_g - \text{FPR}|$, es-AUC $= \text{AUC}/(1 + \sum_g |\text{AUC}_g - \text{AUC}|)$. FairForge-2 SELECT, seed 17.

Method (all Xception)	Fairness metrics (%)									Detection (%)
	g $F_{\text{FPR}}\downarrow$	g $F_{\text{DP}}\downarrow$	g es-AUC↑	r $F_{\text{FPR}}\downarrow$	r $F_{\text{DP}}\downarrow$	r es-AUC↑	i $F_{\text{FPR}}\downarrow$	i $F_{\text{DP}}\downarrow$	i es-AUC↑	AUC↑
Ori (BCE)	3.43	0.11	99.74	19.54	1.52	99.22	40.52	1.62	98.61	99.80
DAG-FDD WACV'24	4.28	0.18	99.66	18.32	1.51	99.26	37.89	1.68	98.50	99.81
DAW-FDD WACV'24	3.43	**0.09**	**99.80**	12.87	1.29	99.38	30.70	1.56	98.85	**99.84**
PG-FDD CVPR'24	– pending (not run on FF2) –									
F3Net (FAD branch)	– pending (trained; verification failed) –									
CORE	– pending (trained; verification failed) –									
Ding et al. CVPR'26 (corrected GDA, our run)	**0.72**	0.24	98.98	**4.09**	1.38	97.21	**12.44**	1.72	95.31	99.06
CES-FL P: original-only BCE	3.60	0.24	**99.80**	12.16	1.06	**99.44**	28.39	1.53	**99.08**	99.81
CES-FL A: weighted BCE	3.26	0.14	99.79	12.80	1.12	99.43	29.10	1.43	99.05	99.81
CES-FL B: A + soft error disparity	3.43	0.19	**99.80**	13.17	1.11	99.42	29.55	**1.32**	99.04	99.81
L_A: plain-head BCE (matched control)	4.11	0.20	99.77	17.54	1.33	99.22	37.37	1.58	98.66	99.81
L_H-PS (ours): hierarchical-real + probe	4.28	0.27	99.67	17.55	1.30	99.02	36.23	1.50	98.27	99.75
RF-S (ours): real-only FPR equalisation, balanced reals	4.11	0.20	99.77	16.02	1.26	99.21	34.12	1.49	98.64	99.80
RF-SJg (ours): real-only FPR eq. + gentle JPEG aug	3.26	0.23	99.74	13.05	**0.83**	99.16	27.53	**1.32**	98.58	99.75
CES-FL full (ours, final)	– pending (not run on FF2) –									

Table 3: Axes beyond Ding et al. (values ×100). Age: <30, 30–44, 45–59, 60+. Skin: 4 ITA bins (unknown excluded). Race F_{FPR} within source recomputes the race metric inside one source dataset, against that source's own FPR (FairFace: 4 races, 2,196 reals; FFHQ: 3 races, 242 reals). FNR = overall miss rate.

Method (all Xception)	Age (4 bins)			Skin tone (4 bins)			Race $F_{\text{FPR}}\downarrow$ within source		
	$F_{\text{FPR}}\downarrow$	es-AUC↑	worst AUC↑	$F_{\text{FPR}}\downarrow$	es-AUC↑	worst AUC↑	FairFace	FFHQ	FNR↓
Ori (BCE)	8.74	99.61	99.69	14.57	99.50	99.66	1.63	4.19	0.30
DAG-FDD WACV'24	10.45	99.58	99.72	12.78	99.40	99.69	2.58	11.76	**0.22**
DAW-FDD WACV'24	10.45	99.59	99.72	**6.80**	**99.62**	**99.73**	2.43	15.14	0.24
PG-FDD CVPR'24	– pending (not run on FF2) –								
F3Net (FAD branch)	– pending (trained; verification failed) –								
CORE	– pending (trained; verification failed) –								
Ding et al. CVPR'26 (corrected GDA, our run)	**1.58**	98.41	98.69	7.72	97.46	98.29	4.88	**2.07**	1.07
CES-FL P: original-only BCE	9.23	99.63	99.70	11.68	**99.62**	99.70	1.64	18.15	0.38
CES-FL A: weighted BCE	9.02	99.57	99.67	10.62	99.61	99.71	1.41	10.35	0.33
CES-FL B: A + soft error disparity	7.42	99.60	99.68	11.56	**99.62**	99.69	1.86	10.35	0.37
L_A: plain-head BCE (matched control)	9.58	**99.69**	99.73	14.23	99.53	99.67	1.82	8.72	0.28
L_H-PS (ours): hierarchical-real + probe	11.90	99.40	99.60	13.35	99.30	99.61	1.46	9.63	0.38
RF-S (ours): real-only FPR equalisation, balanced reals	9.62	**99.69**	**99.74**	13.38	99.53	99.68	2.27	9.24	0.29
RF-SJg (ours): real-only FPR eq. + gentle JPEG aug	8.97	99.63	99.70	9.72	99.43	99.59	**1.29**	11.85	0.45
CES-FL full (ours, final)	– pending (not run on FF2) –								

Table 4: False-positive rate (%) on real images per gender×race group (lower is better; the overall target is 10%). Real images per group in brackets. FairForge-2 SELECT, seed 17.

Method (all Xception)	Female Asian (202)	Female Black (194)	Female White (257)	Female Others (348)	Male Asian (322)	Male Black (238)	Male White (256)	Male Others (621)
Ori (BCE)	15.84	18.56	15.18	**3.74**	11.18	13.87	12.89	3.38
DAG-FDD WACV'24	16.34	18.56	14.01	5.75	9.94	13.45	14.06	**2.90**
DAW-FDD WACV'24	13.37	15.46	14.79	7.18	**7.14**	14.71	12.11	5.48
PG-FDD CVPR'24	– pending (not run on FF2) –							
F3Net (FAD branch)	– pending (trained; verification failed) –							
CORE	– pending (trained; verification failed) –							
Ding et al. CVPR'26 (corrected GDA, our run)	**12.38**	**11.34**	**8.17**	10.34	8.39	13.03	**8.59**	9.50
CES-FL P: original-only BCE	17.33	11.86	15.56	6.61	8.70	13.03	11.33	5.48
CES-FL A: weighted BCE	15.84	12.37	17.12	5.46	9.01	13.03	10.94	5.80
CES-FL B: A + soft error disparity	15.35	13.40	16.34	6.03	9.01	13.03	11.72	5.31
L_A: plain-head BCE (matched control)	16.83	18.56	15.18	4.31	9.94	11.34	13.67	4.03
L_H-PS (ours): hierarchical-real + probe	15.84	17.01	15.95	5.46	10.25	12.61	13.28	3.38
RF-S (ours): real-only FPR equalisation, balanced reals	15.84	18.56	15.18	4.89	9.94	10.92	12.89	4.51
RF-SJg (ours): real-only FPR eq. + gentle JPEG aug	14.85	14.95	15.56	5.75	10.25	**10.50**	12.50	5.48
CES-FL full (ours, final)	– pending (not run on FF2) –							